Strategic Investment in a Forward-Looking New Keynesian Monetary–Fiscal Game: A Recursive Australian Application

Monetary–Fiscal Coordination Project

1 October 2026

Abstract

This paper studies monetary–fiscal coordination in a forward-looking New Keynesian policy game. The monetary and fiscal authorities act simultaneously each period under different, fixed mandates, while a separate loss represents true social welfare. Private agents understand the stationary policy rules, so expected future policy affects current demand and inflation. Instrument adjustment costs make current policy a predetermined state: a policymaker can alter the position inherited by its future self and, through the rival's feedback rule, alter future rival policy. This is the strategic-investment mechanism. We derive a rational-expectations Markov-perfect equilibrium as a fixed point between the private forecasting rule and coupled policy Riccati equations. An Australian quarterly calibration compares flexible Nash, monetary persistence only, fiscal persistence only, bilateral persistence, and discretionary coordination. Expectations materially change shock transmission and the policy mix, particularly after a cost-push shock. The narrow inherited- policy strategic channel is nevertheless small in the baseline: its bounded contribution is 0.003 for unilateral monetary persistence and 0.002 for unilateral fiscal persistence. Seven dispersed initial forecasting rules converge to one stable linear equilibrium. Forward-looking parameter maps locate economically large policy displacement only when strong transmission and high adjustment persistence coincide, at the cost of a thinner stability margin; isolated large strategic shares can instead reflect cancellation in a small continuation condition. Forward-looking behaviour is therefore a conduit for strategic interaction, not by itself a guarantee of endogenous leadership or equilibrium multiplicity.

Keywords: monetary–fiscal coordination; strategic investment; New Keynesian model; Markov-perfect equilibrium; rational expectations; adjustment costs; Australia.
JEL codes: C73, E52, E58, E61, E62.

1 Introduction

Monetary and fiscal authorities affect the same output gap, inflation rate, and public balance sheet. Their objectives need not be identical. A central bank can place relatively high weight on inflation, a fiscal authority can place relatively high weight on activity, and society can evaluate both using a third, fixed welfare criterion. These differences create policy conflict, but conflict alone does not imply that one authority can manipulate the other or that several equilibria exist. Strategic leadership requires a way for a current action to change a payoff-relevant future state and a rival response to that state.

Adjustment costs provide precisely such a state. If changing the policy rate is costly, today's monetary choice becomes tomorrow's inherited policy position. A monetary authority that knows the fiscal feedback rule can take account of the fiscal response to that position. The same logic applies to a fiscal instrument. With an adjustment cost for only one authority, that authority has a

unilateral commitment capacity. With costs for both, both authorities invest in inherited positions. The resulting allocation is a simultaneous Markov-perfect equilibrium, not a mechanically imposed Stackelberg order. It can become leader-like only when the induced rival responses are sufficiently strong.

The private economy must also be forward looking. Households and firms care about expected future activity, inflation, and policy. A promise of future stabilisation can therefore change current outcomes before the promised action is taken. This adds a second strategic conduit: current policy changes the state, the state changes expected future policy, and those expectations feed back into current demand and price setting. The central question is whether this rational-expectations feedback strengthens the strategic-investment mechanism enough to create endogenous leadership or difficult equilibrium selection.

We answer that question in a compact quarterly New Keynesian model calibrated to Australian data. Monetary and fiscal instruments enter the dynamic IS curve, inflation follows a forward-looking Phillips curve, and public debt is predetermined. Each authority solves an infinite-horizon linear–quadratic problem under discretion. Private forecasts and policy rules are jointly determined. The exercise compares five institutional environments:

1. flexible feedback Nash, with no adjustment cost;
2. monetary persistence only;
3. fiscal persistence only;
4. bilateral persistence; and
5. coordination under a separate true social objective with flexible instruments.

The distinction between the last case and commitment is essential. Coordination changes whose objective is minimized; adjustment costs change the intertemporal technology of policy; rational expectations determine how the private sector prices future policy. A coordinated discretionary planner is not a Ramsey planner that can bind all future decisions.

The calibrated results are informative in part because they do not force a large leadership effect. Policy inertia changes the feedback rules and shock responses. Forward-looking private behaviour is visible, especially after a cost-push shock. But the rival's coefficient on the other authority's inherited instrument is small, so the specifically Petkov-style strategic-investment component is small. This separates three ideas that are often run together: mandate conflict, ordinary dynamic spillovers through debt and expectations, and strategic investment through an inherited policy position.

2 Connection to the literature

The private block follows the standard forward-looking New Keynesian logic of Clarida, Galí and Gertler (1999): expected future activity and expected future inflation affect current choices. Unlike a conventional Taylor-rule exercise, both policy instruments are endogenous solutions to agency optimization problems. The model is therefore a dynamic monetary–fiscal game in the tradition of Oudiz and Sachs (1984), Tabellini (1986), and Beetsma and Bovenberg (1997).

The strategic-investment interpretation follows Petkov (2004) and the dynamic oligopoly analysis of Jun and Vives (2004). A durable or costly-to- adjust action changes tomorrow's state and hence a rival's feedback action. The important object is not adjustment cost in isolation, but the

product of state exposure, the rival's response coefficient, and the shadow value of the rival action. Coury and Petkov (2010) applies related dynamic-incentive logic to monetary delegation and credibility.

Forward-looking policy games can also possess several self-consistent discretionary equilibria. Kirsanova and Yates (2025) show how private expectations can transmit strategic complementarity across current and future policymakers when an endogenous persistent state is present. Their result clarifies what must be tested here: a rational-expectations fixed point can be multiple, but forward-looking equations do not make it multiple by construction. The direction of monetary–fiscal interaction can likewise depend on shock type, transmission, mandates, and policy inertia (Muscatelli, Tirelli and Trecroci, 2004).

3 The forward-looking economy

3.1 Private behaviour

Let x_t be the output gap, π_t the inflation gap, m_t the monetary instrument, and f_t the fiscal instrument. Positive m_t is contractionary in demand; positive f_t is expansionary. Private behaviour is

$$x_t = \mathbb{E}_t x_{t+1} - \sigma\big(m_t - \mathbb{E}_t \pi_{t+1}\big) + \psi f_t + d_t, \tag{1}$$

$$\pi_t = \beta_p \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t, \tag{2}$$

where d_t is a demand disturbance and u_t a cost-push disturbance. The IS curve says that current demand rises with expected future activity, a lower expected real interest rate, and fiscal expansion. The Phillips curve says that current inflation combines expected future inflation, marginal-cost pressure represented by x_t, and a cost-push shock.

These equations matter strategically because future policy rules enter $\mathbb{E}_t x_{t+1}$ and $\mathbb{E}_t \pi_{t+1}$. For example, anticipated future monetary tightening lowers expected future demand; anticipated future fiscal support can raise it. Current policy therefore has both a direct effect and an expectations-mediated effect.

The public-debt gap evolves according to

$$b_{t+1} = \rho_b b_t + \chi_m m_t + \chi_f f_t - \chi_\pi \pi_t. \tag{3}$$

This is a parsimonious law of motion rather than a full nominal government valuation equation. Monetary tightening raises debt-service pressure, fiscal expansion raises the debt gap, and inflation erodes it.

For the baseline experiment, the innovations are one-period impulses, $d_{t+1} = \rho_d d_t$ and $u_{t+1} = \rho_u u_t$ with $\rho_d = \rho_u = 0$. Endogenous persistence remains because debt and, where applicable, lagged instruments are predetermined.

3.2 Timing and the Markov state

At the start of period t, the authorities and private agents observe

$$s_t = (b_t, m_{t-1}, f_{t-1}, d_t, u_t)'. \tag{4}$$

The monetary and fiscal authorities then choose m_t and f_t simultaneously as functions of s_t. Private agents observe those choices, know the stationary policy rules that will be used from $t+1$ onward,

and determine x_t and π_t from equations (1)–(2). Finally, equation (3) and the policy choices determine s_{t+1}.

The simultaneous within-period policy move is deliberate. Leadership is not imposed by allowing one authority to move first. It can emerge only through the state transition: a current action changes an inherited position to which future policymakers respond.

The Julia implementation uses a seven-element state vector by retaining two zero-coefficient bookkeeping locations for lagged output and inflation. In the forward-looking model they have no economic role; the minimal state is the five-element vector above. This implementation detail does not add backward- looking terms to equations (1) or (2).

3.3 Agency mandates and true social welfare

The monetary authority minimizes

$$J_M = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_M^t \left[\omega_\pi^M \pi_t^2 + \omega_x^M x_t^2 + \omega_b^M b_{t+1}^2 + r_M m_t^2 + \lambda_M (m_t - m_{t-1})^2\right], \tag{5}$$

and the fiscal authority minimizes

$$J_F = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_F^t \left[\omega_\pi^F \pi_t^2 + \omega_x^F x_t^2 + \omega_b^F b_{t+1}^2 + r_F f_t^2 + \lambda_F (f_t - f_{t-1})^2\right]. \tag{6}$$

The monetary mandate is relatively inflation-oriented and the fiscal mandate relatively output-oriented. Each agency bears only the resource or implementation costs of its own instrument. All weights are primitives held fixed across regimes.

True social welfare is represented by the separate loss

$$\begin{aligned} J_S = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_S^t \big[&\omega_\pi^S \pi_t^2 + \omega_x^S x_t^2 + \omega_b^S b_{t+1}^2 + r_M^S m_t^2 + r_F^S f_t^2 \\ &+ \lambda_M (m_t - m_{t-1})^2 + \lambda_F (f_t - f_{t-1})^2\big]. \end{aligned} \tag{7}$$

This formulation distinguishes delegated objectives from social welfare. Cooperation means that a single discretionary policymaker chooses both instruments to minimize (7); it does not average the two agency losses or change the primitive weights between experiments.

The adjustment coefficients have two interpretations that should not be confused. They are real implementation costs in the loss functions, and they also create a commitment technology because the current instrument becomes a payoff-relevant state. In the experiments below the coefficients are imposed institutional primitives, not choices in an outer game.

4 Recursive rational-expectations equilibrium

4.1 Reducing the private block

The equilibrium is linear. Let $y_t = (x_t, \pi_t)'$ and $q_t = (s_t', m_t, f_t)'$. Before imposing private equilibrium, write the state transition as

$$s_{t+1} = N q_t + E y_t. \tag{8}$$

Debt and the inherited instruments are contained in N; current output and inflation enter the next state through E. Suppose private agents conjecture the stationary forecasting rule

$$\mathbb{E}_t y_{t+1} = G s_{t+1}. \tag{9}$$

Equations (1)–(2) can then be collected as

$$Ly_t = Dq_t + HGs_{t+1}, \quad L = \begin{bmatrix} 1 & 0 \\ -\kappa & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & \sigma \\ 0 & \beta_p \end{bmatrix}. \tag{10}$$

Substituting (8) gives

$$M(G)y_t = [D + HGN]q_t, \qquad M(G) = L - HGE, \tag{11}$$

$$y_t = O(G)q_t, \qquad O(G) = M(G)^{-1}[D + HGN], \tag{12}$$

$$s_{t+1} = T(G)q_t, \qquad T(G) = N + EO(G). \tag{13}$$

The inverse $M(G)^{-1}$ is the within-period expectations multiplier. It summarizes how beliefs about the next state feed back into current private decisions. Partition

$$T(G) = \begin{bmatrix} A(G) & B_M(G) & B_F(G) \end{bmatrix}. \tag{14}$$

For a fixed G, the policy problem is now an ordinary linear–quadratic dynamic game.

4.2 Coupled policy Bellman equations

Let the stationary feedback rules be

$$m_t = k_M' s_t, \qquad f_t = k_F' s_t. \tag{15}$$

For a fixed expectations map, each stage loss can be written $q_t' Q_i(G) q_t$. The matrices $Q_i(G)$ include current output and inflation because those variables have already been mapped into q_t by $O(G)$. Let $V_i(s) = s' P_i s$. The two simultaneous first-order conditions are

$$\begin{bmatrix} Q^M_{mm} + \beta_M B_M' P_M B_M & Q^M_{mf} + \beta_M B_M' P_M B_F \\ Q^F_{fm} + \beta_F B_F' P_F B_M & Q^F_{ff} + \beta_F B_F' P_F B_F \end{bmatrix} \begin{bmatrix} m_t \\ f_t \end{bmatrix} = - \begin{bmatrix} Q^M_{ms} + \beta_M B_M' P_M A \\ Q^F_{fs} + \beta_F B_F' P_F A \end{bmatrix} s_t. \tag{16}$$

Solving (16) produces k_M and k_F. Define $K = \begin{bmatrix} k_M' \\ k_F' \end{bmatrix}$ as the two-row policy matrix and $C = \begin{bmatrix} I \\ K \end{bmatrix}$ as the map from states into $(s_t', m_t, f_t)'$. The closed-loop law and value updates are

$$A_c = A + B_M k_M' + B_F k_F', \tag{17}$$

$$P_i = C' Q_i C + \beta_i A_c' P_i A_c, \qquad i \in \{M, F\}. \tag{18}$$

Equations (16)–(18) are iterated to solve the feedback Nash game conditional on G. Coordination instead uses the two-instrument first-order condition and a single Riccati equation associated with $Q_S(G)$.

4.3 The expectations fixed point

The conjecture in (9) must agree with the outcomes generated by the policy rules. Partition $O(G) = (O_s, O_m, O_f)$. Rational expectations require

$$G = \Phi(G) \equiv O_s(G) + O_m(G) k_M(G)' + O_f(G) k_F(G)'. \tag{19}$$

The numerical algorithm therefore has two loops. The inner loop solves the coupled policy Riccati equations for a provisional G. The outer loop updates G toward $\Phi(G)$ with damping and repeats until the forecasting residual is below tolerance. The bilateral baseline takes 278 outer iterations and has closed-loop spectral radius 0.7834.

Equilibrium definition. A stationary linear rational-expectations Markov-perfect Nash equilibrium is a tuple (G, k_M, k_F, P_M, P_F) such that: (i) each authority's rule is optimal given the rival's rule and G; (ii) the value matrices satisfy (18); (iii) private forecasts satisfy (19); and (iv) the closed-loop transition is stable. This is a local linear equilibrium concept. It neither presumes open-loop commitment nor imposes a leader.

4.4 Multiplicity and determinacy

There are two distinct stability questions. First, a candidate equilibrium must have a stable economic law of motion, $\rho(A_c) < 1$. Second, the nonlinear mapping $G \mapsto \Phi(G)$ can have one or several fixed points. A fixed point can be economically stable even if a particular undamped numerical iteration does not converge to it, so multi-start computation is evidence rather than a global uniqueness proof.

Local multiplicity becomes more plausible when the derivative $D\Phi(G)$ has an eigenvalue near or above one. Economically, that can occur when future policy reacts strongly to the state, the private expectations multiplier $M(G)^{-1}H$ is large, and the state itself is persistent. Then an expectation of aggressive future policy changes today's economy enough to validate that policy response. Weak feedback slopes or heavily penalized instruments break this loop.

5 Where strategic investment enters

5.1 Ordinary strategic interaction versus strategic investment

Both authorities always interact because both instruments affect x_t, π_t, and b_{t+1}. This is ordinary within-period strategic interaction. Debt adds an ordinary dynamic channel: one authority changes debt and the other later responds to debt. Forward-looking private behaviour adds an expectations channel: anticipated future responses affect x_t and π_t today.

Strategic investment is narrower. Suppose authority i chooses a_{it} and the rival's next-period rule is $a_{j,t+1} = k_j' s_{t+1}$. A marginal current action changes the rival according to

$$\frac{\partial a_{j,t+1}}{\partial a_{it}} = k_j' B_i. \tag{20}$$

If a_{it} becomes the inherited-policy component of s_{t+1}, the corresponding term in (20) is the rival's coefficient on that lagged instrument. Current policy is then an investment in a future strategic position.

The continuation first-order condition can be decomposed schematically as

$$\underbrace{\beta_i B_i' P_i s_{t+1}}_{\text{total continuation effect}} = \underbrace{\mathcal{N}_i}_{\text{own-state effect}} + \underbrace{\beta_i \sum_h B_{ih} k_{jh} \mathcal{H}^i_{j,t+1}}_{\text{rival-response effect}}, \tag{21}$$

where $\mathcal{H}^i_{j,t+1}$ is authority i's shadow loss from a marginal future rival action. The code reports the rival-response term and separates states h into macro states and the inherited own-policy state. The latter is the direct empirical counterpart of the Petkov mechanism.

5.2 Why an adjustment cost creates commitment capacity

When $\lambda_M = 0$, m_{t-1} does not enter the monetary loss. In the present model it then has no economic role, and neither authority reacts to it. With $\lambda_M > 0$, tomorrow's monetary authority wants to

avoid a large movement away from m_t, so m_t affects m_{t+1}. If fiscal policy reacts to m_t as an inherited state, today's monetary authority internalizes that fiscal reaction. Thus adjustment costs do not simply smooth a time series; they change the strategic state space.

A useful scalar approximation illustrates why the effect need not rise monotonically with λ_i. Ignoring cross-effects, the coefficient on an inherited instrument behaves roughly like

$$k_{i,a_{i,-1}} \approx \frac{\lambda_i}{r_i + \lambda_i + \beta_i B_i' P_i B_i}. \tag{22}$$

At zero cost there is no inherited-policy channel. A moderate cost makes policy persistent and can strengthen strategic investment. At a very large cost, however, the authority can barely move the current instrument, so its ability to invest in a different future position collapses. The economically relevant effect is generally hump-shaped rather than "the larger the cost, the stronger the leader."

5.3 Parameter combinations that make the motive important

Equation (21) identifies six complementary ingredients:

1. *State exposure:* B_i must map current policy into a persistent state, either directly through the inherited instrument or indirectly through debt.
2. *Rival sensitivity:* $k_j' B_i$ must be large. Different mandate weights can steepen this response, but need not do so.
3. *Payoff exposure:* the induced rival action must materially affect authority i's inflation, output, debt, or instrument loss.
4. *Patience and persistence:* high β_i and persistent endogenous states raise the value of changing the future path.
5. *Expectations transmission:* a large $\|M(G)^{-1}H\|$ allows anticipated future policies to affect current outcomes strongly.
6. *Feasible movement:* adjustment costs must be large enough to make policy inherited but not so large that current investment is prohibitively expensive.

These conditions are multiplicative. Strongly divergent objectives with a weak fiscal multiplier can produce little strategic investment. Strong private expectations with almost no rival response can change shock propagation but not create leadership. Conversely, a persistent balance-sheet state and steep cross-policy rules can create significant strategic effects even when objective weights differ only modestly.

5.4 Forward-looking parameter regions

To make the preceding conditions quantitative, we solve two parameter-region experiments using the fully forward-looking model in every cell. There is no mixing with a backward-looking specification. The first experiment varies the strength of direct policy transmission and the degree of adjustment-induced persistence. Specifically, a common transmission scale τ multiplies both σ and ψ, preserving their baseline ratio, while adjustment costs are represented by the dimensionless share

$$p_i = \frac{\lambda_i}{D_i + \lambda_i}, \tag{23}$$

where D_i is authority i's one-period own-action curvature in the flexible Nash equilibrium at that value of τ. Thus p_i measures the fraction of local policy curvature contributed by adjustment. It is comparable across transmission calibrations even though the raw λ_i is not. The Australian values correspond to $p_M = 0.330$ and $p_F = 0.282$.

The strategic statistic in the top panels of Figure 1 is

$$S_i^{\text{inv}} = \frac{|\text{inherited-policy rival-response term}|}{|\text{inherited-policy term}| + |\text{remainder of the continuation FOC}|}. \tag{24}$$

It is bounded between zero and one and isolates the adjustment-state component of equation (21). The lower-left panel provides an economic-scale check: it adds the distances between each unilateral-persistence policy pair and flexible Nash after the demand shock. The lower-right panel reports $1 - \rho(A_c)$. A light region in a share panel is persuasive evidence of an important strategic motive only if policy displacement is also material and the stability margin remains acceptable.

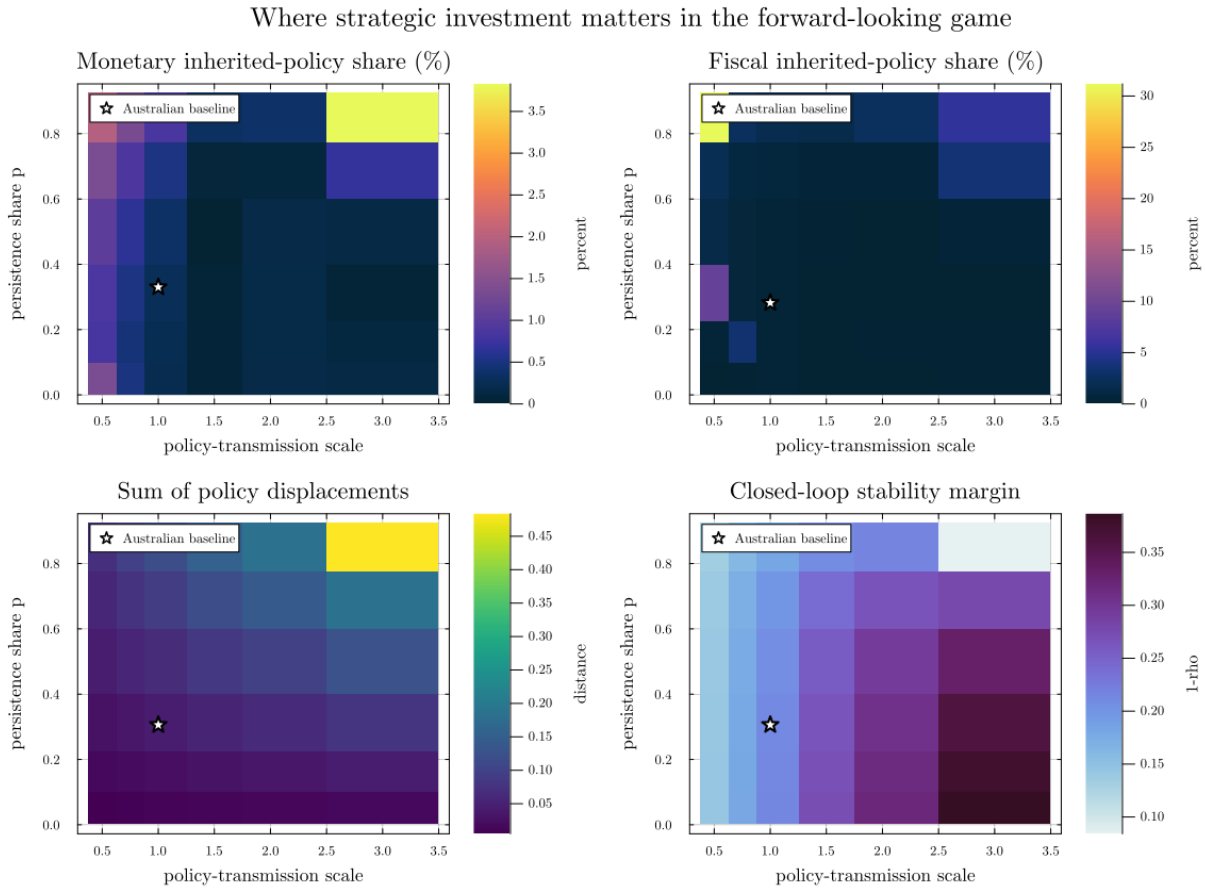


Figure 1: Strategic-investment regions in the fully forward-looking game. The two upper panels use authority-specific colour scales. Stars show the Australian persistence share; the common star in the lower panels uses their average only as a visual reference.

All 36 of 36 cells converge to stable rational-expectations solutions. Three findings stand out.

First, the Australian calibration lies in a dark region: modest persistence and baseline transmission produce a small inherited-policy contribution. This confirms the local decomposition in Figure 5 using a much wider parameter neighborhood.

Second, the economically largest displacement occurs when both ingredients are strong. At $\tau = 3.00$ and $p = 0.85$, the sum of unilateral policy displacements reaches 0.484. The stability margin falls to 0.084, however. Strong policy transmission makes a current investment consequential, while very high persistence makes it durable; the same combination moves the equilibrium closer to a fragile closed-loop boundary. The monetary inherited-policy share reaches 3.83 percent in this high-transmission, high-persistence corner.

Third, a large *share* need not mean a large aggregate strategic effect. The fiscal share reaches 31.19 percent at weak transmission $\tau = 0.50$ and high persistence $p = 0.85$. Yet the associated total policy displacement is only 0.071, with stability margin 0.134. In that cell the non-investment continuation terms nearly cancel, so a modest inherited-policy term occupies a large fraction of a small denominator. This is why the paper reports shares, policy distances, and stability together.

Expectations and debt persistence. The second experiment holds the Australian adjustment coefficients fixed and varies the private discount factor β_p and debt persistence ρ_b. This separates the strength of the forward-looking Phillips-curve channel from the durability of the main macroeconomic state. Figure 2 reports the largest monetary or fiscal inherited- policy share, the bilateral policy displacement from flexible Nash, the norm of the private expectations multiplier $M(G)^{-1}H$, and the stability margin.

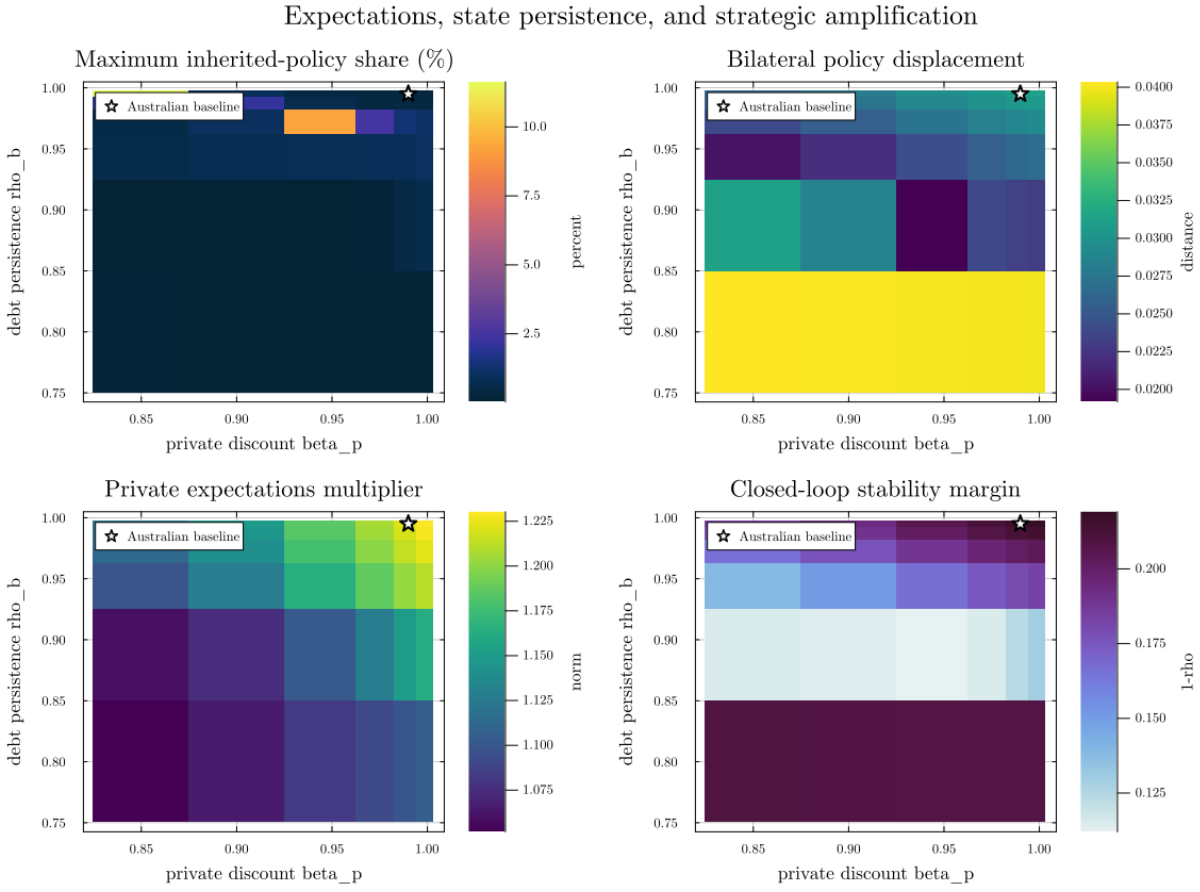


Figure 2: Private expectations, debt persistence, and strategic effects with the Australian bilateral adjustment costs.

All 36 of 36 cells converge and the smallest stability margin is 0.112. As expected, the private expectations multiplier increases with β_p and reaches 1.230 in the grid. But the strategic-investment share does not increase monotonically with that multiplier. Its maximum, 11.63 percent, occurs at $\beta_p = 0.850$ and $\rho_b = 0.995$, where policy displacement is only 0.026. Conversely, the largest bilateral policy displacement, 0.040, occurs at $\beta_p = 0.850$ and $\rho_b = 0.800$, but its inherited-policy share is only 0.10 percent.

This lack of monotonicity is economically meaningful. A larger β_p strengthens the passage from expected future inflation to current inflation, but it can also improve anticipated stabilization and reduce the future shadow loss that an authority wants to manipulate. Greater ρ_b makes current policy effects more durable, but can make both authorities react mainly to the same debt state rather than to one another's inherited instrument. Forward- lookingness and persistence are therefore enabling conditions, not sufficient statistics for strategic investment.

Table 1 summarizes the comparative-static logic. Signs are generally conditional because each parameter changes several pieces of the response product in equation (21).

Table 1: Parameters governing strategic effects in the forward-looking game

Parameter	Main channel	When it raises strategic investment
λ_i or p_i	Makes current policy an inherited state	At intermediate-to-high persistence, provided current movement remains feasible and the rival reacts to the inherited instrument
σ, ψ	Strengthen direct policy transmission to demand	When stronger transmission steepens cross-policy feedback rather than merely improving each authority's own stabilization
κ	Passes output and policy effects into inflation	When the induced inflation effect is important under the acting authority's mandate and does not destabilize expectations
β_p	Strengthens expected-inflation feedback	When the larger private multiplier also raises the shadow value of manipulating future rival policy; not mechanically
ρ_b and χ coefficients	Increase durability and policy exposure of debt	When the rival reacts strongly to debt and that response affects the acting authority's loss
β_M, β_F	Raise the weight on continuation payoffs	More directly increases the value of strategic investment, conditional on a nonzero rival response
Mandate divergence	Changes reaction slopes and desired policy mix	When it makes $k_j' B_i$ steep; disagreement alone is insufficient
r_M, r_F	Penalize instrument levels	Lower penalties permit stronger reactions, but can also reduce stability margins

The robust high-strategy region is consequently an intersection, not a single threshold: policy

must materially affect a persistent state, the rival must respond to that state, private expectations must transmit the future response, and the induced action must matter for the first authority's loss. The heat maps show why changing objective weights or adjustment costs one at a time can fail to generate the game of interest.

5.5 Bilateral persistence and the meaning of "meta-Nash"

With $\lambda_M > 0$ and $\lambda_F > 0$, both current instruments become states. Each authority can attempt to shape its future self and its rival. The stationary solution is a Nash equilibrium in policy rules with bilateral commitment capacity. It is not literally a Nash equilibrium in the adjustment- cost parameters: the authorities do not choose λ_M and λ_F. A true meta-game would add an institutional stage in which the authorities choose delegation or implementation technologies, anticipating the policy equilibrium that follows.

6 Australian calibration and experiments

Table 2 reports the quarterly baseline. The private discount factor is calibrated. The transmission coefficients are descriptive Australian estimates based on quarterly macroeconomic series and should not be read as a fully identified structural NK estimation. The experiment is a transparent quantitative laboratory for strategic mechanisms. Data inputs are drawn from Reserve Bank of Australia statistical tables (Reserve Bank of Australia, 2026).

Table 2: Forward-looking quarterly calibration

Object	Parameter	Value
Intertemporal transmission	σ	0.03695
Fiscal demand multiplier	ψ	0.05142
Phillips-curve slope	κ	0.15511
Private discount factor	β_p	0.99
Debt persistence	ρ_b	0.995
Debt effects	$(\chi_m, \chi_f, \chi_\pi)$	$(0.015, 0.060, 0.100)$
Demand and cost-push persistence	(ρ_d, ρ_u)	$(0, 0)$
Monetary mandate	$(\omega_\pi^M, \omega_x^M, \omega_b^M, r_M)$	$(1.50, 0.25, 0.15, 0.10)$
Fiscal mandate	$(\omega_\pi^F, \omega_x^F, \omega_b^F, r_F)$	$(0.50, 0.75, 0.15, 0.10)$
Social weights	$(\omega_\pi^S, \omega_x^S, \omega_b^S, r_M^S, r_F^S)$	$(1.00, 0.50, 0.15, 0.10, 0.10)$
Agency and social discount factors	$\beta_M, \beta_F, \beta_S$	0.99
Persistence experiments	(λ_M, λ_F)	$(0.05, 0.04)$
Shock sizes	(d_0, u_0)	$(-0.4639, 0.6700)$

The demand experiment begins with a one-standard-deviation contraction. The cost-push experiment begins with a one-standard-deviation inflationary shock. For each shock we solve all five regimes anew, including the private rational- expectations fixed point. Reported "macro loss" evaluates the common social inflation, output, debt, and instrument-level terms while excluding physical adjustment costs. "Actual loss" includes the applicable adjustment costs. This distinction prevents costly implementation from being mistaken for a free commitment device.

7 Results

7.1 Impact policies and welfare

Table 3 reports the demand contraction. Flexible Nash uses $m_0 = -0.0874$ and $f_0 = -0.0843$. Because positive fiscal policy is expansionary, the fiscal response is contractionary despite the negative demand innovation. This is not a claim that fiscal contraction is generically optimal after a demand shortfall. It is a property of this strategic calibration, in which expected future stabilization and debt incentives interact with a transitory shock.

Table 3: Demand contraction: impact policies and discounted social loss

Regime	m_0	f_0	Macro loss	Actual loss
Flexible Nash	-0.0874	-0.0843	0.1154	0.1154
Monetary persistence	-0.0660	-0.0802	0.1149	0.1153
Fiscal persistence	-0.0883	-0.0648	0.1151	0.1154
Bilateral persistence	-0.0667	-0.0615	0.1147	0.1153
Coordination	-0.1369	-0.0540	0.1149	0.1149

Coordination assigns more of the initial stabilization to monetary policy ($m_0 = -0.1369$) and less to fiscal policy ($f_0 = -0.0540$). This is why the coordinated monetary response can be larger than the responses under agency Nash games: the planner internalizes both agencies' macroeconomic spillovers and chooses the least costly joint instrument mix under the social weights. It is not caused by changing those weights across regimes.

Table 4 reports the cost-push experiment. All regimes tighten monetary policy and use a positive fiscal instrument in this calibration. The planner uses a smaller monetary tightening than flexible Nash but more fiscal support. Coordination has the lowest actual social loss, 0.5325. Bilateral persistence reduces the macro component relative to flexible Nash but raises actual loss to 0.5460 once implementation costs are counted.

Table 4: Cost-push inflation: impact policies and discounted social loss

Regime	m_0	f_0	Macro loss	Actual loss
Flexible Nash	0.2516	0.4091	0.5428	0.5428
Monetary persistence	0.2029	0.3989	0.5385	0.5413
Fiscal persistence	0.2606	0.3182	0.5429	0.5476
Bilateral persistence	0.2098	0.3099	0.5385	0.5460
Coordination	0.1934	0.4763	0.5325	0.5325

7.2 How expectations enter on impact

Figure 3 decomposes the two private equations term by term. Every group of bars sums exactly to the realized impact output or inflation gap. In the demand experiment, the innovation itself dominates current output. Expected output and inflation are small because the shock is transitory and policy rapidly stabilizes the induced state. The figure is useful precisely because it prevents a visually large policy difference from being misdescribed as a large expectations channel.

After the cost-push shock, expected future output contributes between roughly 0.03 and 0.05 to current output across regimes. Expected inflation also adds to current inflation. Policy persistence

changes these forecasts, so it changes current private behaviour even before future policy is implemented. Coordination produces a different decomposition because its joint policy rule induces a different forecast map G.

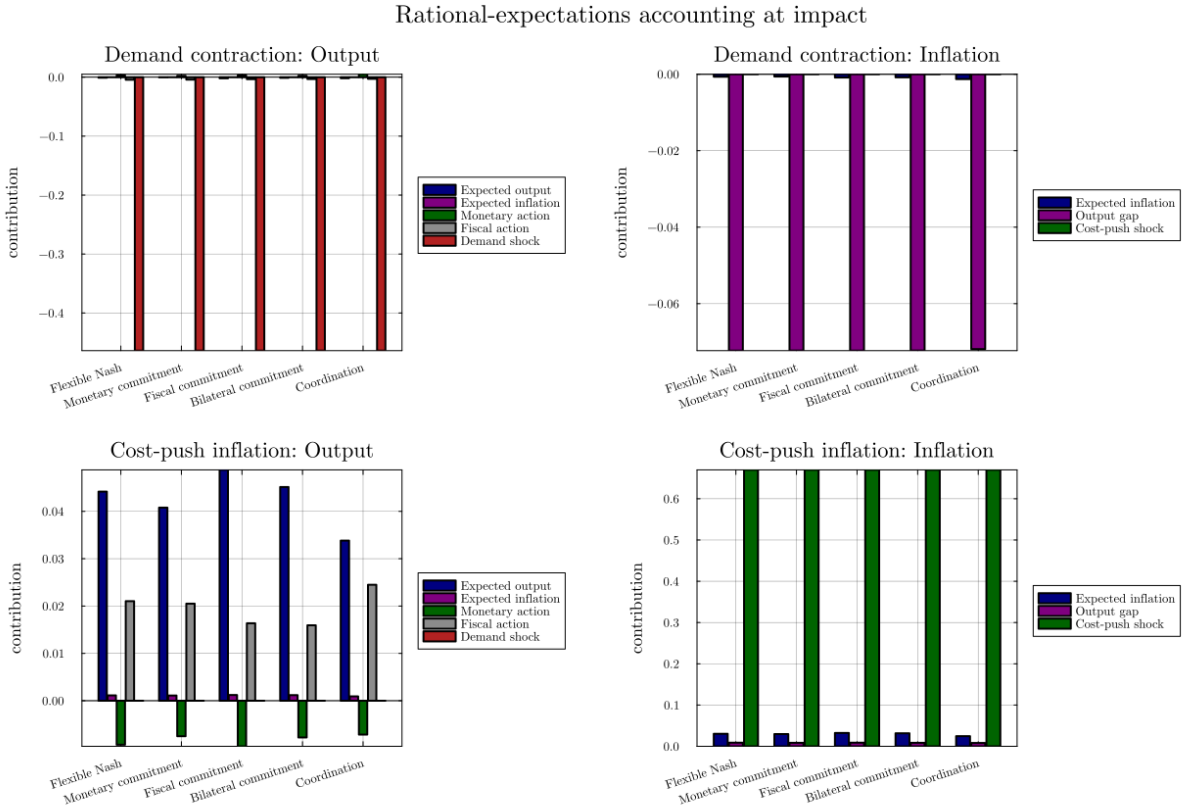


Figure 3: Impact accounting from equations (1) and (2). Bars within each regime sum to the realized variable.

7.3 Feedback rules and the inherited-policy channel

Figure 4 displays the solved feedback coefficients. The blank lagged-output and lagged-inflation columns confirm that the pure forward-looking model does not use those bookkeeping states. Debt is the dominant persistent macro state. Introducing monetary adjustment cost gives the monetary rule a coefficient of about 0.26 on lagged monetary policy. Introducing fiscal adjustment cost gives the fiscal rule a coefficient of about 0.17 on lagged fiscal policy. These own coefficients establish policy persistence.

Leadership, however, depends on the *cross* coefficients. Under monetary persistence the fiscal coefficient on lagged monetary policy is only about -0.03. Under fiscal persistence the monetary coefficient on lagged fiscal policy is about -0.01. Thus current policy substantially binds the authority's future self but only weakly moves the rival. This is the main reason the calibrated model does not generate a large leader advantage.

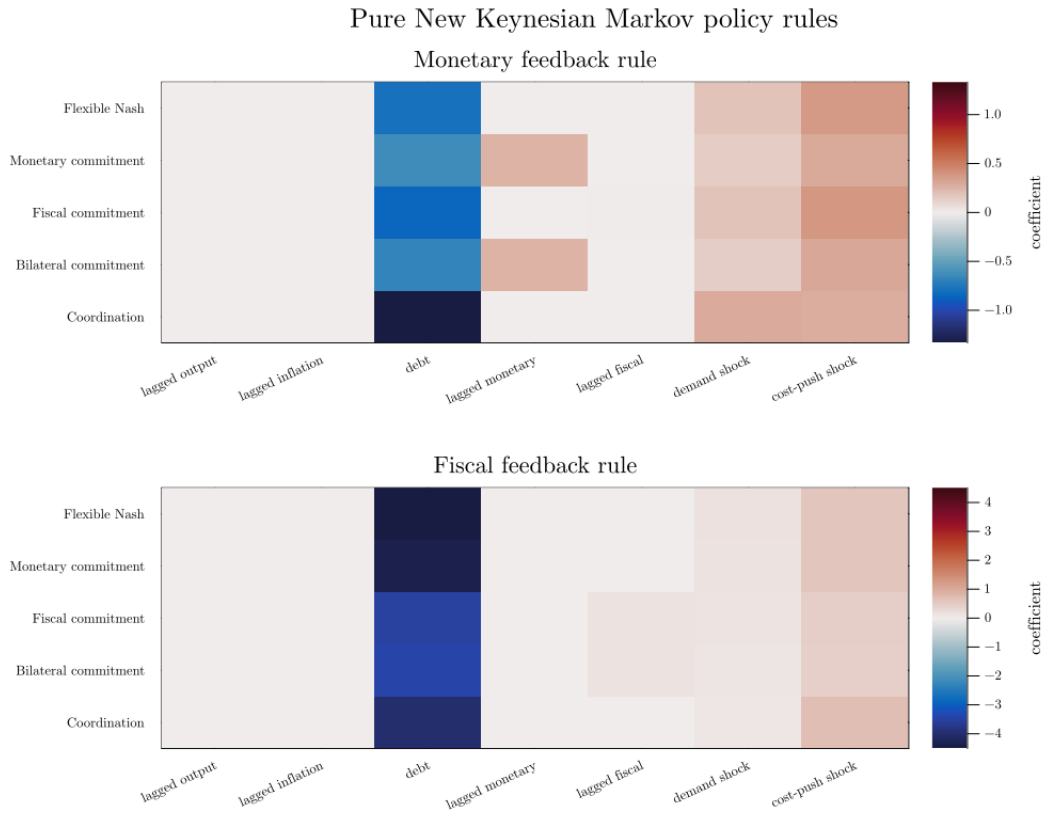


Figure 4: Stationary Markov policy-rule coefficients. Separate colour scales are used for the monetary and fiscal panels.

Figure 5 reports the continuation first-order-condition decomposition after the demand shock. For unilateral monetary persistence, the macro-state contribution is -3.503e-05 and the inherited-policy contribution is -1.281e-05. For unilateral fiscal persistence the corresponding values are -2.269e-05 and -4.864e-06. In bounded-share terms, the direct inherited-policy component is only 0.003 for the monetary authority and 0.002 for the fiscal authority. The signs indicate the direction of the marginal incentive; magnitude is the relevant object for the strength of strategic investment.

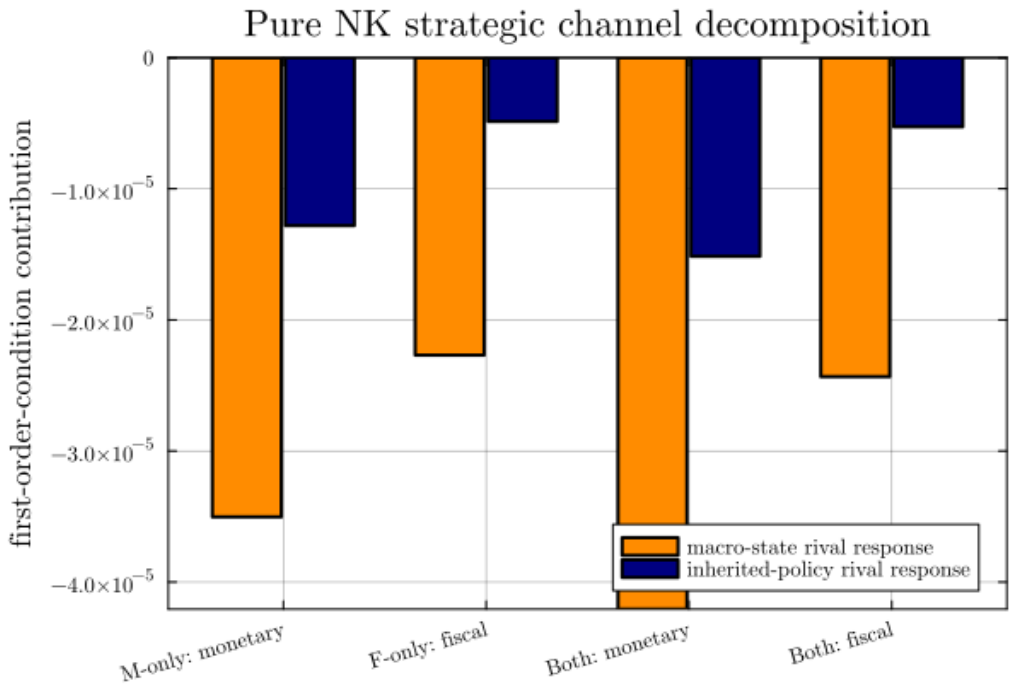


Figure 5: Rival-response contributions to the continuation first-order condition. Orange bars operate through macro states; blue bars isolate the inherited own-policy state that adjustment costs activate.

The conclusion is not that forward-looking behaviour is unimportant. Expectations materially alter the private transmission map and the equilibrium rules. Rather, the particular mechanism needed for endogenous leadership is weak because the rival does not react strongly to the inherited instrument. Expectations amplify whatever future-policy response exists; they cannot create a large $k_j' B_i$ when the policy game produces a small one.

7.4 Dynamic responses

Figures 6 and 7 show the first sixteen quarters. The one-period innovations disappear immediately, but debt and lagged policy carry their effects forward. Under an adjustment-cost regime, the instrument path moves gradually because the current action becomes the next period's reference point. Private agents anticipate the entire path, so the impact output and inflation responses already incorporate those future moves.

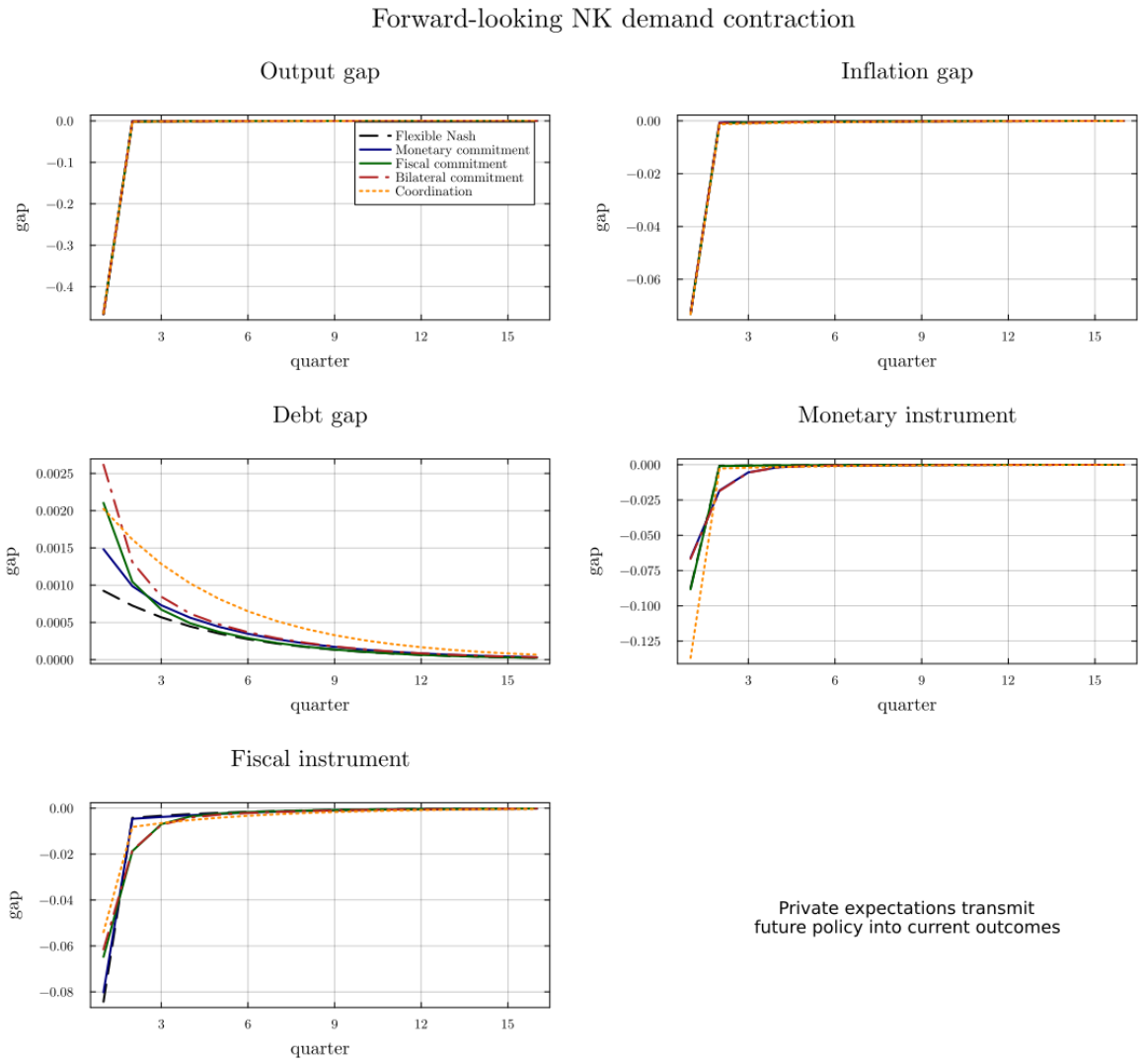


Figure 6: Responses to a one-standard-deviation demand contraction.

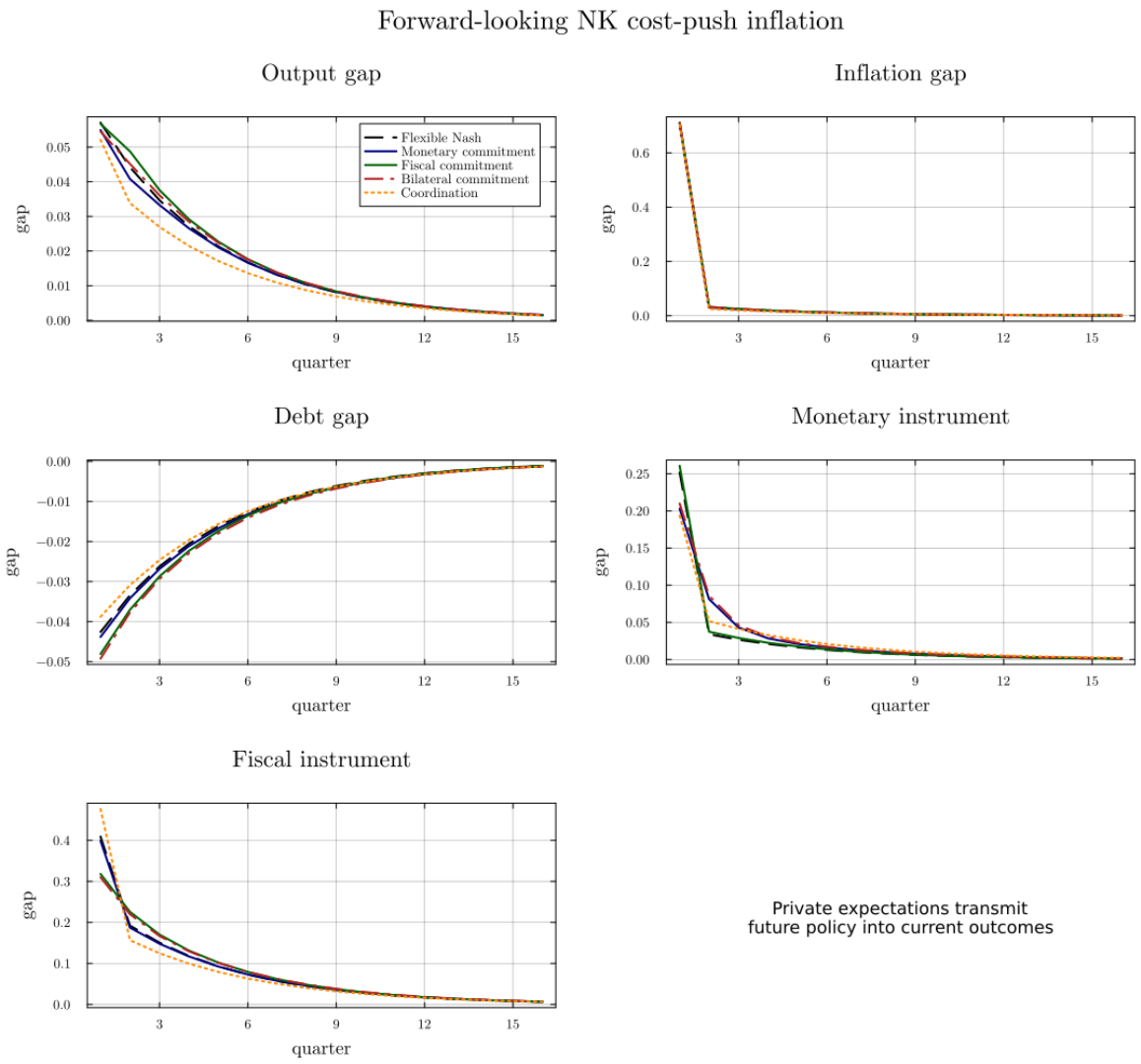


Figure 7: Responses to a one-standard-deviation cost-push shock.

The welfare comparison also shows why coordination and commitment capacity must remain distinct. In the demand experiment bilateral persistence has macro loss 0.1147, slightly below coordinated discretion's 0.1149, because inertia can supply partial history dependence. Its actual loss is nevertheless 0.1153, above the coordinated value 0.1149, after the real cost of moving instruments is included. A restriction can improve macro stabilization while still being socially costly to operate.

7.5 Equilibrium search

The bilateral rational-expectations game was resolved from 7 dispersed initial forecasting maps, including negative, zero, and large positive scalings of a reference map. All converged attempts identified 1 distinct stable stationary solution. This is evidence that equilibrium selection is not difficult in the baseline. It is not a theorem of global uniqueness: nonlinear extensions, a fiscal limit, capital, persistent shocks, or a stronger private expectations multiplier could introduce additional branches.

8 Interpretation and extensions

8.1 What the forward-looking formulation changes

The forward-looking formulation changes the economic question in three ways. First, an announced future path affects current allocations, so a policy investment need not wait one period to have macroeconomic consequences. Second, beliefs about future policy are equilibrium objects. The authorities optimize taking the private forecasting rule as given, while the forecasting rule must be validated by their choices. Third, equilibrium multiplicity can arise from self-confirming expectations even when the policy first-order conditions are locally well behaved.

It does not follow that forward-looking behaviour always magnifies strategic investment. If anticipated stabilization makes the future state small, it can reduce the shadow value of manipulating that state. If the rival's rule is insensitive to the inherited instrument, expectations mainly transmit the authority's own future policy rather than a strategic rival response. And if the closed-loop system approaches instability, a large expectations effect may signal fragile determinacy rather than useful commitment.

8.2 Why different objective weights are not enough

Different mandates determine what the authorities want, but the structural equations determine what they can achieve by manipulating one another. Suppose fiscal policy places much more weight on output than monetary policy. This creates disagreement about the preferred mix of m_t and f_t. Yet if ψ is small, debt responses are weak, and the fiscal rule barely responds to inherited monetary policy, a monetary investment changes little about future fiscal behaviour. The disagreement is real but strategically inoperative.

Conversely, even moderate mandate differences can matter greatly when debt is persistent, the rival's reaction is steep, and expected future policy strongly affects present outcomes. The relevant analytical object is therefore a response product such as

$$\beta_i \times (B_i \text{ state exposure}) \times (k_j \text{ rival slope}) \times (\text{payoff shadow value}) \times (\text{expectations multiplier}). \quad (25)$$

Calibration should target these components, not only the relative inflation and output weights.

8.3 Relation to fiscal dominance

The model contains a fiscal-pressure channel. Fiscal policy changes debt, debt is persistent, inflation changes the real debt burden, and monetary policy responds to debt. It can therefore study how fiscal actions constrain or redirect monetary stabilization.

It does not yet fully nest the fiscal theory of the price level or a strong fiscal-dominance regime. That would require nominal government liabilities, a consolidated intertemporal government budget constraint, and explicit rules for monetary versus fiscal backing. The price level would then help satisfy a nominal valuation equation. Adding that block would be the natural extension for asking whether adjustment-cost commitment and private expectations alter the boundary between monetary and fiscal dominance.

8.4 Most informative next experiments

The baseline points to targeted extensions rather than an indiscriminate increase in adjustment costs:

1. estimate the cross-policy feedback slopes and the fiscal demand multiplier using Australian data;
2. allow persistent demand and cost-push processes, which raise the value of shaping future policy;
3. add nominal long-term government debt and a government valuation equation;
4. introduce capital or another private endogenous state that can support stronger intertemporal complementarity;
5. solve a Ramsey benchmark to separate coordination under discretion from full commitment; and
6. if "meta-Nash" is to be literal, add an outer institutional game in which commitment capacities are chosen rather than imposed.

9 Conclusion

This paper formulates the monetary–fiscal problem directly as a forward- looking New Keynesian recursive game. The authorities choose policy simultaneously each period, private agents form model-consistent expectations, and adjustment costs turn current instruments into strategic states. The solution is a fixed point between private forecasts and coupled policy Bellman equations.

The framework clarifies the mechanism. Mandate differences create conflict. Debt and private expectations create dynamic spillovers. Adjustment costs create inherited policy positions. Endogenous leadership requires all of these pieces to produce a sufficiently large rival response and payoff effect. In the Australian baseline, policy becomes persistent and expectations alter current outcomes, but the rival's response to the inherited instrument is weak. The strategic-investment motive is consequently present but small, and a multi-start search finds one stable stationary rational-expectations equilibrium.

This negative quantitative result is useful. It says that adjustment costs are a credible route to strategic commitment, but not a guarantee of a leader-like allocation. The next calibration should discipline the response product identified by the analytical decomposition and then test whether additional persistent private or fiscal states create the complementarities needed for difficult equilibrium coordination.

A Numerical algorithm

For each institutional regime, the Julia solver performs the following steps:

1. initialize the private forecasting matrix G;
2. construct $M(G)$, $O(G)$, and the reduced state matrices $A(G), B_M(G), B_F(G)$;
3. construct the agency loss matrices $Q_M(G)$ and $Q_F(G)$;
4. iterate equations (16)–(18) until the two feedback rules and value matrices converge;
5. compute $\Phi(G)$ from equation (19);
6. update G with damping and repeat until $\|\Phi(G) - G\|_\infty$ is below tolerance; and

7. verify $\rho(A_c) < 1$ and the private equations at off-equilibrium test states.

The coordinated solution replaces steps 3–4 with one social loss matrix and a two-instrument Riccati problem. It still requires the same private rational- expectations fixed point because the planner acts under discretion.

B Reproduction and checks

All reported calculations, tables, and figures are produced in Julia. From the project folder:

```
julia --project=. -e "using Pkg; Pkg.instantiate(; update_registry=false)"
julia --project=. run_nk_extension.jl
julia --project=. run_nk_parameter_regions.jl
julia --project=. runtests_nk.jl
julia --project=. build_tex.jl
```

The first command solves the forward-looking game and writes machine-readable results and figures to `output/`. The test suite verifies both private equations, the rational-expectations fixed point, Bellman and policy residuals, closed-loop stability, adjustment-cost ownership, the strategic-channel decomposition, and multi-start agreement. The build command compiles this source into `FORWARD_LOOKING_NK_POLICY_GAME.pdf`.

These checks establish internal consistency of the linear solution. They do not establish structural identification of the Australian parameters or global uniqueness beyond the class of stationary linear equilibria searched.

References

Beetsma, R. M. W. J. and A. L. Bovenberg (1997). Central bank independence and public debt policy. *Journal of Economic Dynamics and Control* 21, 873–894. https://doi.org/10.1016/S0165-1889(97)00003-1.

Clarida, R., J. Galí and M. Gertler (1999). The science of monetary policy: a New Keynesian perspective. *Journal of Economic Literature* 37, 1661–1707. https://doi.org/10.1257/jel.37.4.1661.

Coury, T. and V. Petkov (2010). Monetary delegation: credibility through dynamic incentives. Dubai Initiative Working Paper, Harvard Kennedy School. https://www.belfercenter.org/publication/monetary-delegation-credibility-through-dynamic-incentives.

Jun, B. and X. Vives (2004). Strategic incentives in dynamic duopoly. *Journal of Economic Theory* 116, 249–281. https://doi.org/10.1016/j.jet.2003.08.005.

Kirsanova, T. and A. Yates (2025). Multiple equilibria in the absence of commitment. *Oxford Review of Economic Policy* 41, 404–425. https://doi.org/10.1093/oxrep/graf022.

Muscatelli, V. A., P. Tirelli and C. Trecroci (2004). Fiscal and monetary policy interactions: empirical evidence and optimal policy using a structural New-Keynesian model. *Journal of Macroeconomics* 26, 257–280. https://doi.org/10.1016/j.jmacro.2003.11.014.

Oudiz, G. and J. D. Sachs (1984). International policy coordination in dynamic macroeconomic models. NBER Working Paper 1417. https://doi.org/10.3386/w1417.

Petkov, V. (2004). Delegated management in dynamic oligopolies. Econometric Society 2004 Australasian Meetings, paper 79. https://ideas.repec.org/p/ecm/ausm04/79.html.

Reserve Bank of Australia (2026). Statistical tables. https://www.rba.gov.au/statistics/tables/.

Tabellini, G. (1986). Money, debt and deficits in a dynamic game. *Journal of Economic Dynamics and Control* 10, 427–442. https://doi.org/10.1016/S0165-1889(86)80001-X.